

ECSE-4040

Digital Electronics

Lecture 14

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Lecture Outline

- Interconnect RC Ladder Modelling
- Interconnect Capacitive Coupling

DEI Course Calendar

Text: D.A. Neamen, *Microelectronics: Circuit Analysis and Design*, 4th Ed., McGraw-Hill, 2010.

Wk	Date	Topic	Readings	Assignments
Wk 1	1/6	Lecture: Tu, Introduction; F, MOSFETs.	Neamen: pp. 1141 – 1144; LTspice Web Site (www.linear.com/ltspice).	Studio: W, No Class.
Wk 2	1/13	Lecture: Tu, MOSFETs; F, MOSFETs.	Neamen: Ch 3. pp. 125 – 146; HJS 3 rd Ed: Ch. 2 pp. 35 – 64.	Studio: W, LT SPICE Tutorial; P2.1, 2.5 (HJS 3 rd Ed.)
Wk 3	1/20	Lecture: Tu, MOSFETs (<i>Parameters, Models</i>); F, MOS Gates (<i>VTC, Noise Margins</i>).	Neamen: Ch. 4 pp. 205 – 215; HJS 3 rd Ed: Ch. 3 pp. 111 – 138.	Studio: W, Parameter extraction, P3.5 (HJS 3 rd Ed.).
Wk 4	1/27	Lecture: Tu, MOS Gates (<i>Inverters</i>); F, MOS Gates (<i>Sizing, VTC, NAND, NOR</i>).	Neamen: Ch. 16 pp. 1147 – 1202; HJS 3 rd Ed: Ch. 4 pp 142 – 180, Ch. 5 pp. 195 – 205.	Studio: W, P4.1 & 4.2 (HJS 3 rd Ed.).
Wk 5	2/3	Lecture: Tu, MOS Gates (<i>RC Delays, Combinational Logic</i>); F, MOS Gates (<i>Noise Sources, NM-Delay, Sequential Logic</i>).	HJS 3 rd Ed: Ch. 4 pp 181 – 186, Ch. 5 pp. 206 – 227.	Studio: W, P4.10 & 4.13 (HJS 3 rd Ed.). <u>MID-TERM EXAM STARTS</u> (at the end of Studio).
Wk 6	2/10	Lecture: Tu, MOS Gates (<i>Sequential Logic, Power</i>); F, MOS Gates (<i>Power</i>)	HJS 3 rd Ed: Ch. 5 pp. 228 – 242.	Studio: W, P4.14, 5.1, & 5.7 (HJS 3 rd Ed.).
Wk 7	2/17	Lecture: Tu, No Class ; F, MOS Gates (<i>Propagation Delay, Capacitance</i>)	HJS 3 rd Ed: Ch. 2 pp. 79 – 83, Ch. 6 pp. 249 – 267.	Studio: W, P5.12 & 5.16 (HJS 3 rd Ed.).
Wk 8	2/24	Lecture: Tu, MOS Gates (<i>Input Slope, Elmore Delay</i>); F, Interconnects.	HJS 3 rd Ed: Ch. 6 pp. 267 – 277; Ch. 3 pp. 100 – 103, Ch. 10 pp. 441 – 468.	Studio: W, P6.4 (HJS 3 rd Ed) <u>MID-TERM EXAM DUE</u>
Wk 9	3/10	Lecture: Tu, BJT (<i>Ebers-Moll Model</i>); F, BJT Inverter (<i>VTC</i>).	Neamen: Ch. 5 pp. 285 – 326; HJ 2 nd Ed: Ch. 5 pp. 108 – 124, Ch. 6 pp. 126 – 147.	Studio: W, P5.1, 5.2, & P6.1 – 6.4 (HJ 2 nd Ed.).
Wk 10	3/17	Lecture: Tu, BJT Gates (<i>ECL</i>); BJT Gates (<i>Low Power ECL</i>).	Neamen: Ch. 17 pp. 1256 – 1276; HJ 2 nd Ed: Ch. 7 pp. 173 – 179, Ch. 8 pp. 192 – 194.	Studio: W, P7.9– 7.12 (HJ 2 nd Ed.).
Wk 11	3/24	Lecture: Tu, BJT Gates (<i>CML</i>); Regenerative Ckts (<i>Mono & Astable</i>).	HJ 2 nd Ed: Ch. 8 pp. 202 – 209.	Studio: W, P8.1 & 8.2 (HJ 2 nd Ed).
Wk 12	3/31	Lecture: Tu, Memories (<i>SRAM</i>); F, Memories (<i>DRAM, Sense Amplifiers</i>)	Neamen: Ch. 16 pp. 1208 – 1221; HJS 3 rd Ed: Ch. 8 pp. 368 – 377; WH 4 th Ed: Ch. 12 pp. 498 – 526	Studio: W, P7.5--7.8.
Wk 13	4/7	Lecture: Tu, Memories (<i>TG Logic</i>); F, Memories (<i>SDRAM, Peripherals</i>)	HJS 3 rd Ed: Ch. 7 pp. 309 – 333, Ch. 8 pp. 359 – 368, 377 – 394	Studio: W
Wk 14	4/14	Lecture: Tu, F (<i>D/A & A/D Converters</i>)	Neamen: Ch. 16 pp. 1226 – 1235	Studio: W, No Class
Wk 15	4/21	Lecture: Tu, Review, Th, F, Study Days		Studio: W, In-Class Final FINAL EXAM DUE

Midterm Exam.

In-Class

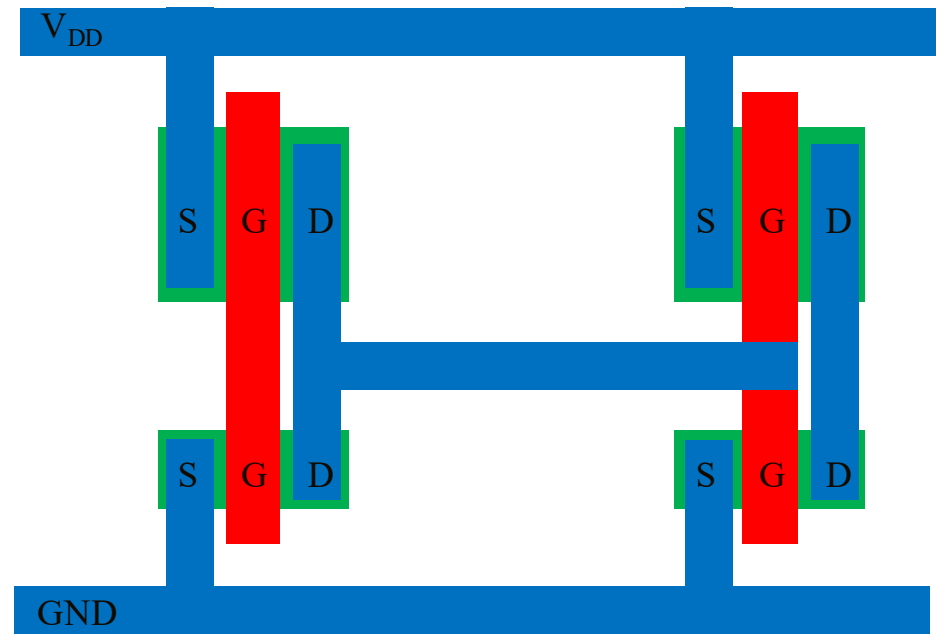
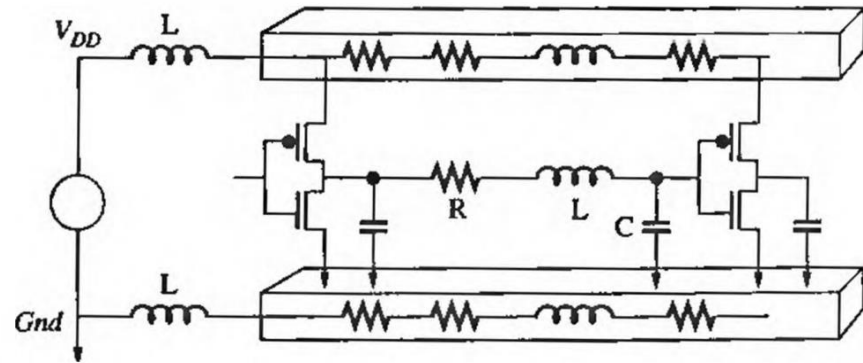
- Taken in Studio session on Feb. 26

Take-Home

- Must be handed in on-time
- Due on Friday (Feb. 28)
- Will allow for as late as midnight submission

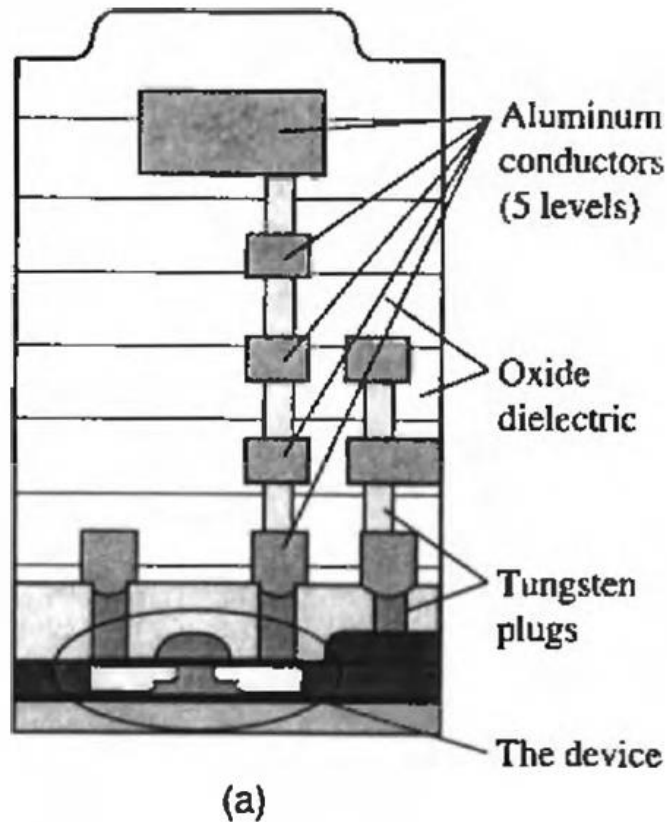
Interconnect Basics

- Logic gates inputs and outputs must be connected together.
- Additional metal adds capacitance
 - To fixed sources
 - GND
 - V_{DD}
 - To variable sources
 - Other signals
 - Clocks
- Longer interconnects have substantial resistance
- τ_p increases at L^2 rate

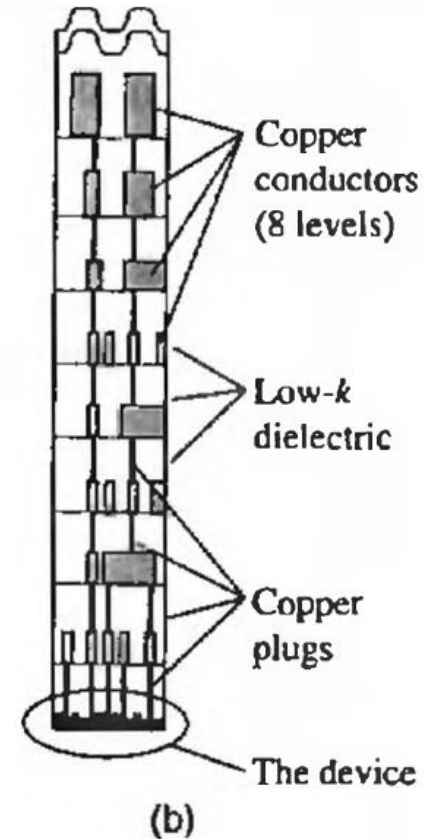


130nm vs. 180nm Interconnects

0.18 μm 5-layer Al metal process

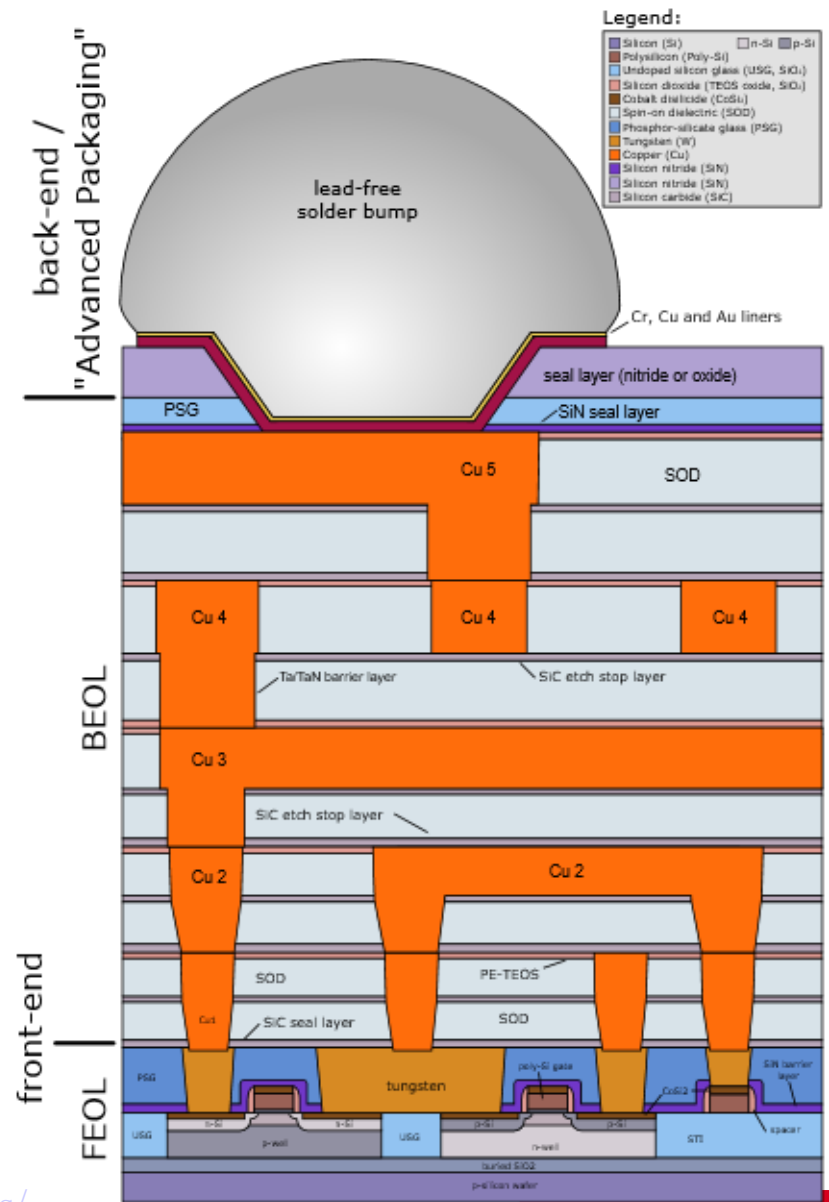
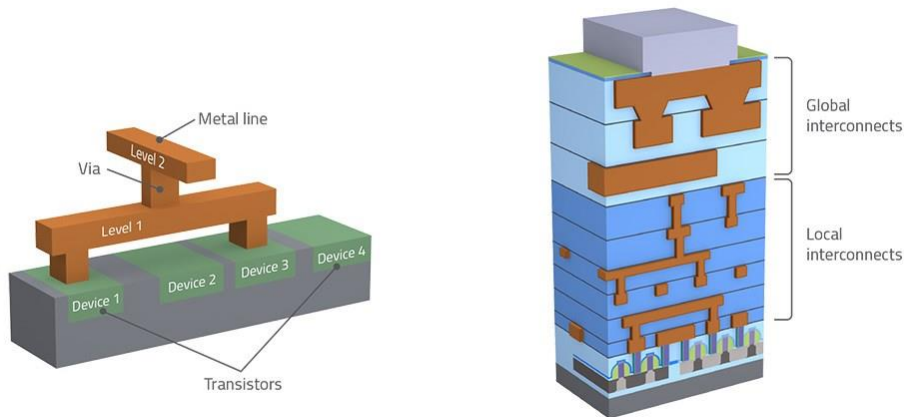


0.13 μm 8-layer Cu metal process



Interconnect Stack

- Most thickness is from metals
 - Copper
 - 5 layers.
 - 12+ on modern process
- Resistivity consistent across layer
 - But not between layers



<https://semiengineering.com/all-about-interconnects/>

https://en.wikipedia.org/wiki/Back_end_of_line

Design Choices

- Designer can choose width, length, & spacing

- $W1, S1, L1, L2$

- Cannot choose height & thickness

- $H1, T1, H2, T2$

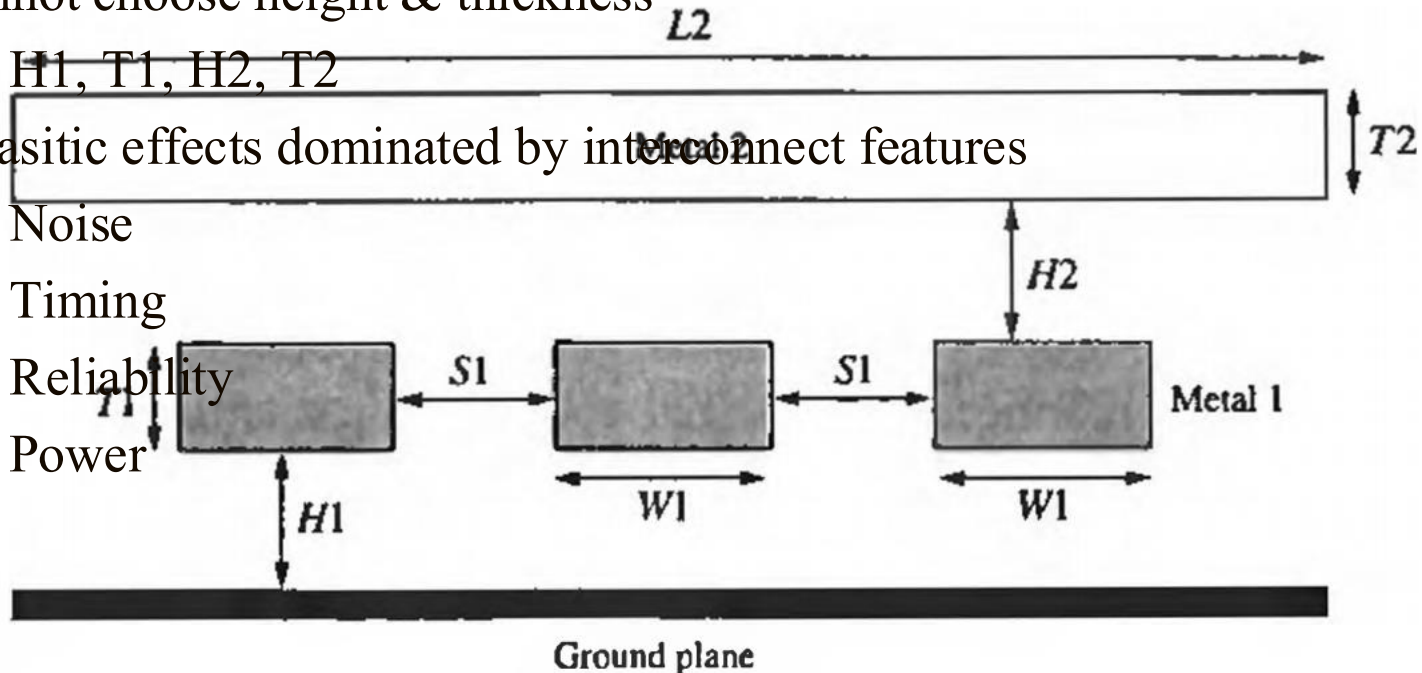
- Parasitic effects dominated by interconnect features

- Noise

- Timing

- Reliability

- Power



Wire Resistance

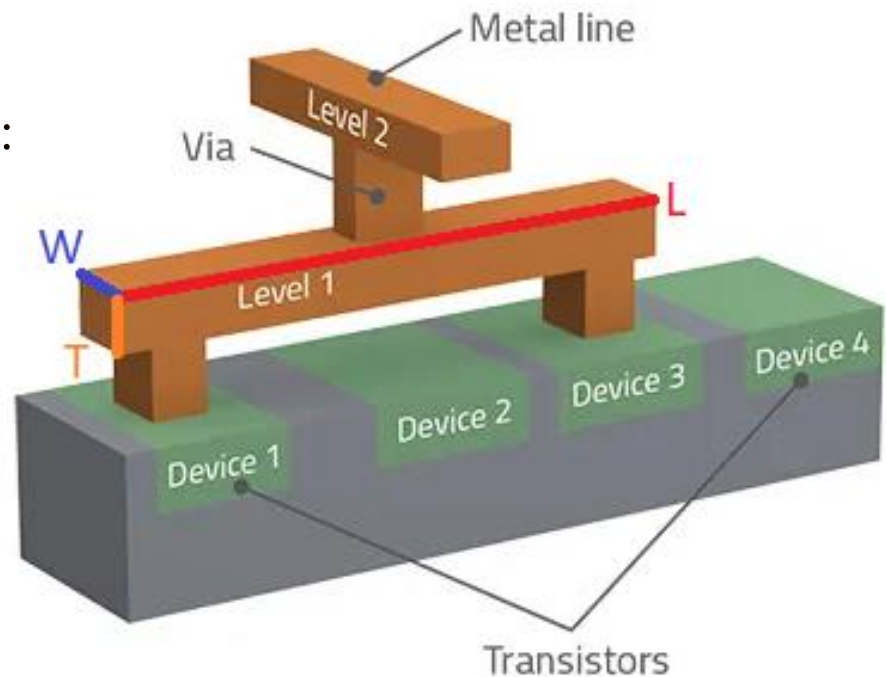
- Designer given R_{sq} (sheet resistance) parameter for each routing layer:

$$R_{sq} = \frac{\rho}{T} \quad [\Omega/]$$

- Designer selects Length & Width:

$$R = \frac{\rho L}{A} = \frac{\rho L}{TW}$$

$$R = R_{sq} \frac{L}{W} [\Omega]$$



Resistance Calculations

- Metals 1-6 are “local” interconnects
 - Designed to be as dense (small) as possible.
- Metals 7-8 are “global” interconnects
 - Designed thicker for lower resistance

$$R_{sq} = \frac{\rho}{T} = \frac{1.7 \mu\Omega\text{-cm}}{0.4 \mu\text{m}} = 42 \text{ m}\Omega/\square \quad \text{Cu} \quad \text{metals 1-6}$$

$$R_{sq} = \frac{\rho}{T} = \frac{1.7 \mu\Omega\text{-cm}}{0.8 \mu\text{m}} = 21 \text{ m}\Omega/\square \quad \text{Cu} \quad \text{metals 7-8}$$

Given these parameters, the actual resistance of a segment of wire is:

$$R = R_{sq} \left(\frac{L}{W} \right)$$

Resistance Calculations

- Ex: 1300 μm long, 130 nm wide interconnect:

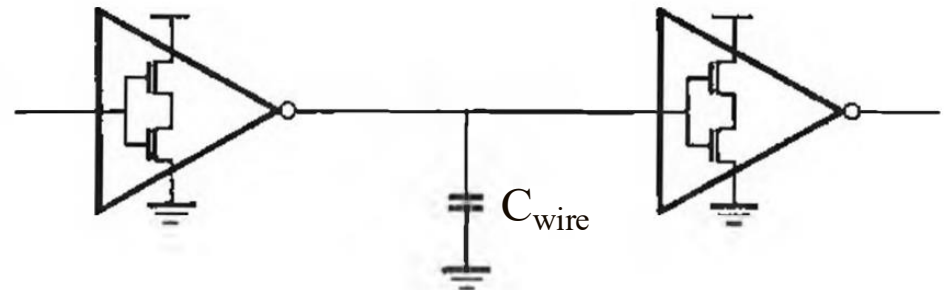
$$R = R_{sq} \frac{L}{W}$$

$$R = 42m \frac{1300000}{130}$$

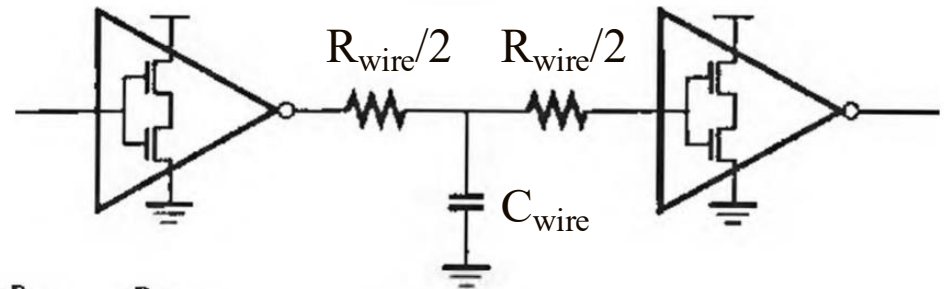
$$R = 420 \Omega$$

RC Ladder

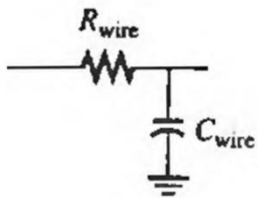
- When interconnect is not short ($> 10\mu\text{m}$), cannot use lumped C model
- Must use RC ladder model



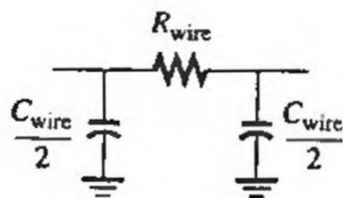
(a)



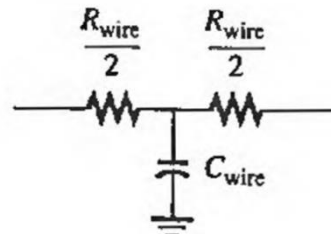
(b)



L-model



Π -model



T-model

INV-wire example

Problem:

Consider an isolated metal 5 Al wire that is 20 mm long and driven by a 100X inverter (i.e., 100 times larger than the minimum size inverter). Since the wire is isolated, its capacitance is $0.1 \text{ fF}/\mu\text{m}$. Estimate the total line resistance and the total line capacitance assuming that the width of the wire is $0.5 \mu\text{m}$. What is the propagation delay as calculated by Elmore for the π -model from the input of the inverter to the end of the wire?

Isolated or loosely packed ($30\text{-}40\lambda$) vs. closely packed ($3\text{-}4\lambda$) interconnecting wires:

$$C_{\text{int}} = 0.1 \text{ fF}/\mu\text{m} \quad \text{instead of normal} \quad C_{\text{int}} = 0.2 \text{ fF}/\mu\text{m}$$

INV-wire example

$$R_{sq} = \frac{\rho}{T} = \frac{2.7 \mu\Omega\text{-cm}}{0.5 \mu\text{m}} = 54 \text{ m}\Omega/\square \quad \text{Al metals 1-4}$$

$$R_{sq} = \frac{\rho}{T} = \frac{2.7 \mu\Omega\text{-cm}}{1.0 \mu\text{m}} = 27 \text{ m}\Omega/\square \quad \text{Al metal 5}$$

$$R_{sq} = \frac{\rho}{T} = \frac{1.7 \mu\Omega\text{-cm}}{0.4 \mu\text{m}} = 42 \text{ m}\Omega/\square \quad \text{Cu metals 1-6}$$

$$R_{sq} = \frac{\rho}{T} = \frac{1.7 \mu\Omega\text{-cm}}{0.8 \mu\text{m}} = 21 \text{ m}\Omega/\square \quad \text{Cu metals 7-8}$$

$$R = R_{sq} \left(\frac{L}{W} \right)$$

INV-wire example

The wire resistance is

$$R_{\text{wire}} = R_{\text{int}}L = \left(\frac{0.027 \Omega/\square}{0.5 \mu\text{m}} \right) (20,000 \mu\text{m}) = 1080 \Omega$$

The wire capacitance is

$$C_{\text{wire}} = C_{\text{int}}L = (0.1 \text{ fF}/\mu\text{m})(20,000 \mu\text{m}) = 2 \text{ pF}$$

The inverter on-resistance is

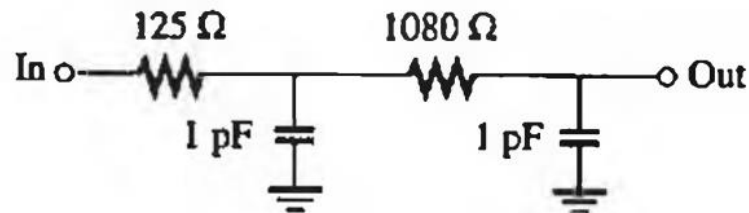
$$R_{\text{eff}} = R_{\text{eqn}}/100 = 12.5 \text{ k}\Omega/100 = 125 \Omega$$

The inverter output capacitance is

$$C_{\text{self}} = C_{\text{eff}}(2W + W)100 = (1 \text{ fF}/\mu\text{m})(0.6 \mu\text{m})100 = 60 \text{ fF}$$

INV-wire example

We can ignore the junction capacitance since it is rather small. Using the π -model, the wire capacitance is split in half while the full wire resistance is used in the model.

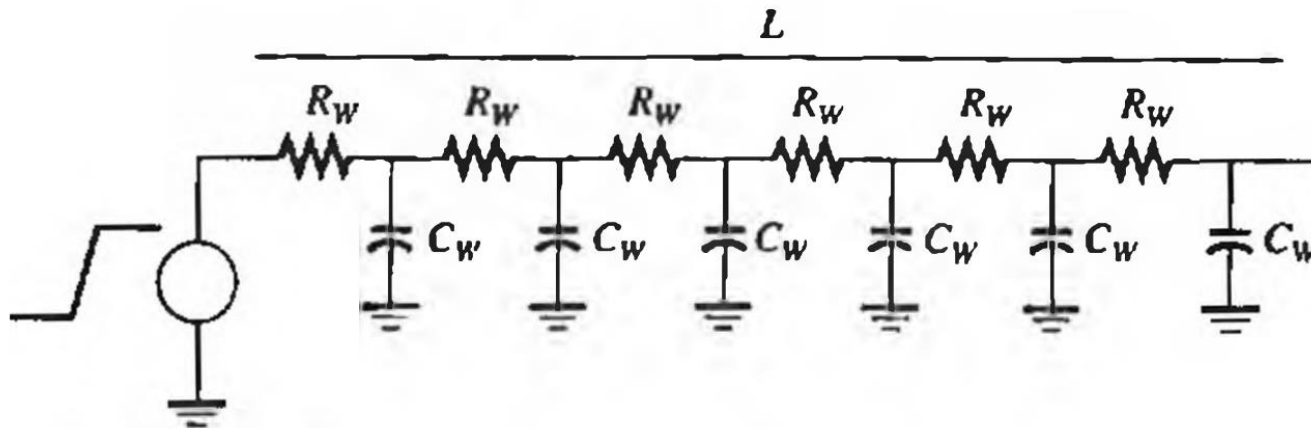


The resulting Elmore delay is

$$\tau_{\text{Elmore}} = (125\ \Omega)(1\ \text{pF}) + (125\ \Omega + 1080\ \Omega)(1\ \text{pF}) \approx 1.3\ \text{ns}$$

RC ladder

- Prior Example broke down capacitance into strict C_{self} , lumped C_{wire} segments.
 - What if we broke it down further (looking at wire only)?
- Lumped-sum approach:
 - $\tau_p = 0.7RC$
 - $\tau_p = 0.7(6R_W)(6C_W)$
 - $\tau_p = 25.2R_WC_W$
- Elmore approach:
 - $\tau_p = 0.7(R_W * C_W + 2R_WC_W + 3R_WC_W + 4R_WC_W + 5R_WC_W + 6R_WC_W)$
 - $\tau_p = 14.7R_WC_W$
 - More accurate for longer segments



RC ladder

- When looking at just the wire:
- Lumped-sum approach:
 - $\tau_p = 25.2R_W C_W$
 - $\tau_p = 25.2 * 180 * 333\text{f}$
 - $\tau_p = 1.51\text{ ns}$
- Elmore approach:
 - $\tau_p = 14.7R_W C_W$
 - $\tau_p = 881\text{ ps}$
- With 6 segments, $\tau_p = 0.583$ lumped RC

RC Ladder – Long Wire

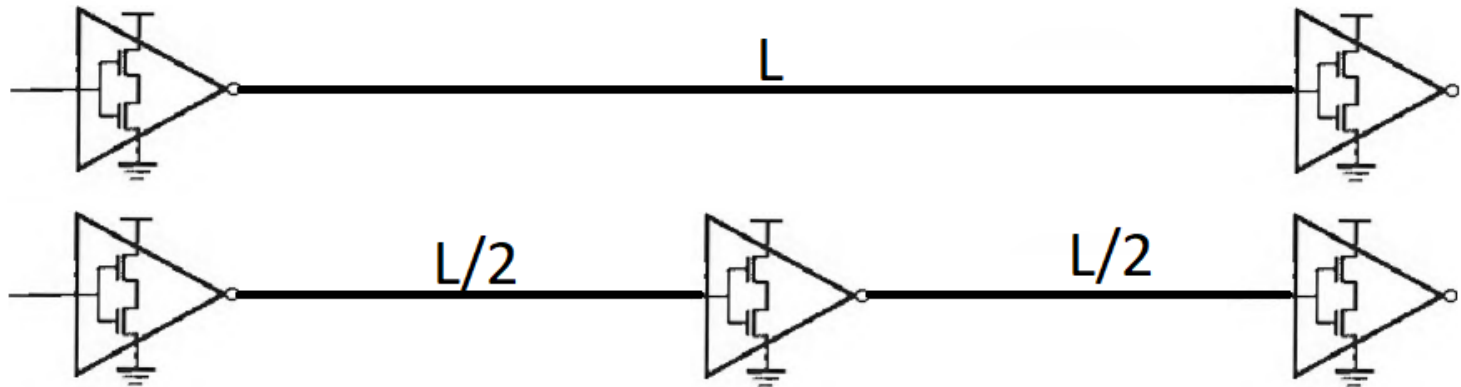
- 6 segments in prior example was arbitrary
 - What is the limit with infinite segments?
- Where $r = R_{sq}$, $c = C_{sq}$:

$$\begin{aligned}\tau_{\text{Elmore}} &= (r\Delta L)(c\Delta L) + 2(r\Delta L)(c\Delta L) + \cdots + n(r\Delta L)(c\Delta L) \\ &= (\Delta L)^2 rc(1 + 2 + \cdots + n) \\ &= (\Delta L)^2 rc(n)(n + 1)/2 \\ &\approx (\Delta L)^2 rcn^2/2 = L^2 rc/2 = R_{\text{wire}} C_{\text{wire}}/2\end{aligned}$$

- With 6 segments, $\tau_p = 0.583$ lumped RC
- With infinite segments, $\tau_p = 0.5$ lumped RC
- Reality: Even Elmore overpredicts delay!
 - $\tau_p = 0.38$ * lumped RC

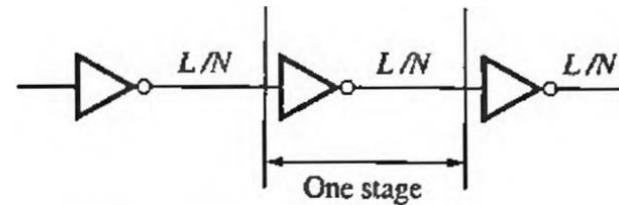
Buffer Insertion

- $\tau_p = 0.7RC$
- If we cut wire into two $L/2$ segments:
 - $C = C/2$
 - $R = R/2$
 - $\tau_p = 2 * 0.7(R/2)(C/2)$
 - $\tau_p = 0.7RC/2$

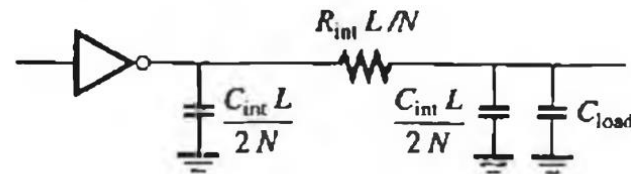


Buffer Insertion

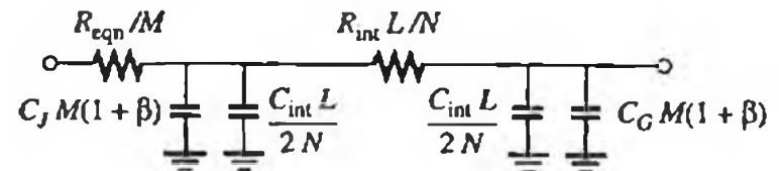
- N number of buffers inserted
- M INV width
- $\beta = W_P/W_N$
 - $1 + \beta = W_P + W_N$
- π model for interconnect



(a) Wire of length L with N buffers inserted



(b) One segment of buffer and interconnect



Buffer Insertion

- Inverter effective resistance:

$$R_{eff} = R_{eqn}/M$$

$$C_{out} = C_{eff}W(1 + \beta)M = C_I(1 + \beta)M$$

$$C_{in} = C_gW(1 + \beta)M = C_G(1 + \beta)M$$

- Total delay:

$$\begin{aligned} t_{total} = N \times t_{segment} = & N(C_G + C_I)R_{eqn}(1 + \beta) \\ & + \left(C_G(1 + \beta)R_{int}M + \frac{C_{int}R_{eqn}}{M} \right)L + \left(\frac{C_{int}R_{int}}{2N} \right)L^2 \end{aligned}$$

Buffer Insertion

- Ideal N and M:

$$\therefore N = \sqrt{\frac{R_{\text{int}} C_{\text{int}} L^2 / 2}{R_{\text{eqn}} (C_J + C_G) (1 + \beta)}} = \sqrt{\frac{t_{\text{wire}}}{\text{FOI}}}$$

$$\therefore M = \sqrt{\frac{R_{\text{eqn}}}{C_G (1 + \beta)} \frac{C_{\text{int}}}{R_{\text{int}}}}$$

Buffer Insertion Example

- Now we optimize this prior problem with buffer insertion:

Problem:

Consider an isolated metal 5 Al wire that is 20 mm long and driven by a 100X inverter (i.e., 100 times larger than the minimum size inverter). Since the wire is isolated, its capacitance is $0.1 \text{ fF}/\mu\text{m}$. Estimate the total line resistance and the total line capacitance assuming that the width of the wire is $0.5 \mu\text{m}$. What is the propagation delay as calculated by Elmore for the π -model from the input of the inverter to the end of the wire?

Using the following values:

$$R_{\text{int}} = 54 \text{ m}\Omega/\mu\text{m} \quad C_{\text{int}} = 0.1 \text{ fF}/\mu\text{m} \quad L = 20,000 \mu\text{m}$$

$$R_{\text{eqn}} = 12.5 \text{ k}\Omega/\square \quad C_g = 2 \text{ fF}/\mu\text{m} \quad C_{\text{eff}} = 1 \text{ fF}/\mu\text{m} \quad W = 0.2 \mu\text{m} \quad \beta = 2$$

Buffer Insertion Example

The number of segments in the wire is

$$N = \sqrt{\frac{R_{\text{int}} C_{\text{int}} L^2 / 2}{R_{\text{eqn}} (C_G + C_I) (1 + \beta)}} = 6.9 \approx 7$$

The buffer size, assuming all equal buffers, is

$$M = \sqrt{\frac{R_{\text{eqn}} C_{\text{int}}}{C_G (1 + \beta) R_{\text{int}}}} \approx 140$$

Then, the buffer resistance is

$$R_{\text{eff}} = \frac{12.5k\ \Omega}{140} \approx 90\ \Omega$$

Buffer Insertion Example

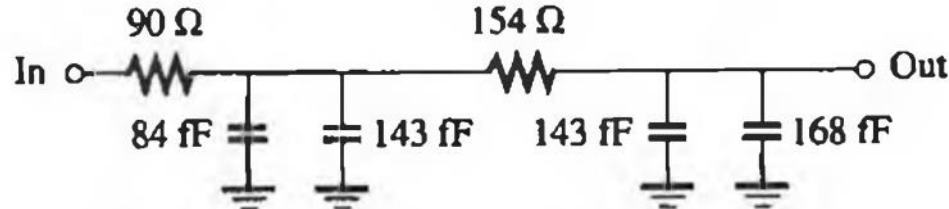
The buffer output capacitance is:

$$C_{\text{self}} = C_{\text{eff}}(2W + W)140 = (1 \text{ fF}/\mu\text{m})(0.6 \mu\text{m})140 = 84 \text{ fF}$$

The buffer input capacitance is

$$C_{\text{fanout}} = C_g(2W + W)140 = (2 \text{ fF}/\mu\text{m})(0.6 \mu\text{m})140 = 168 \text{ fF}$$

Each segment produces this RC delay model.



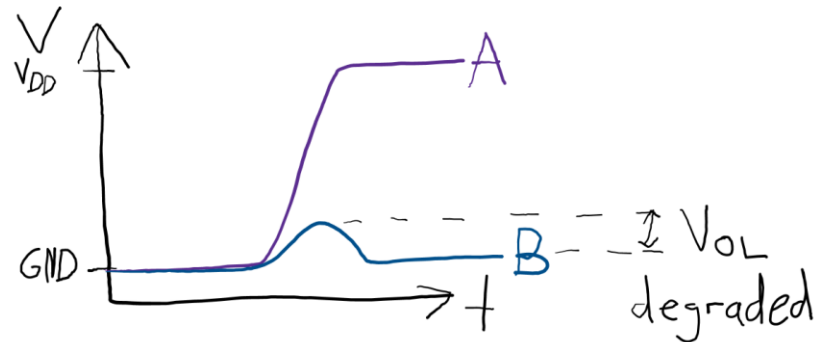
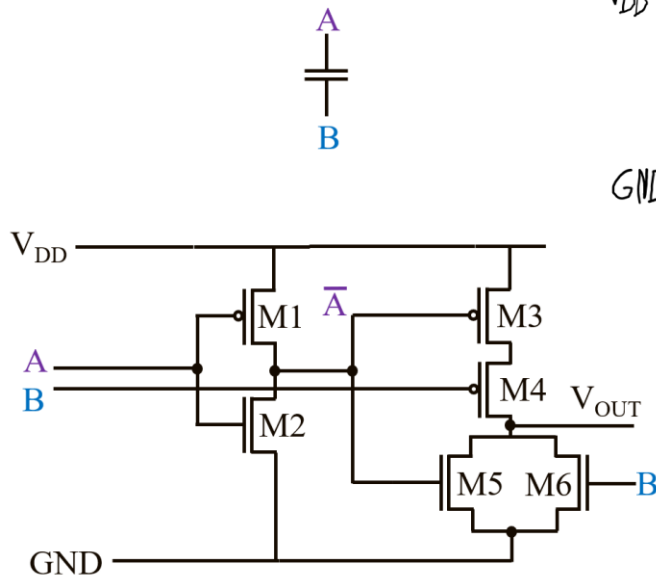
The total Elmore delay is

$$\begin{aligned}\tau_{\text{Elmore}} &= 7 \times [(90 \Omega)(84 \text{ fF} + 143 \text{ fF}) + (90 \Omega + 154 \Omega)(143 \text{ fF} + 168 \text{ fF})] \\ &= 0.67 \text{ ns}\end{aligned}$$

The total delay has been cut in half using buffer insertion.

Capacitive Coupling

- Remember crosstalk noise source:
- If A and B run close together over a “long” distance, they will have a significant capacitance.



C_{coupling} = ε*A/d, d is miniscule

A 

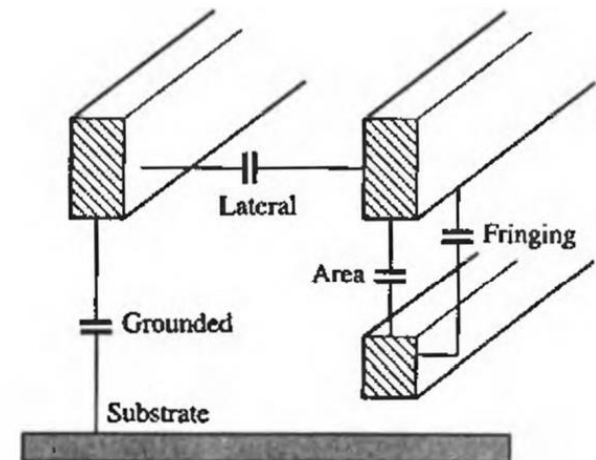
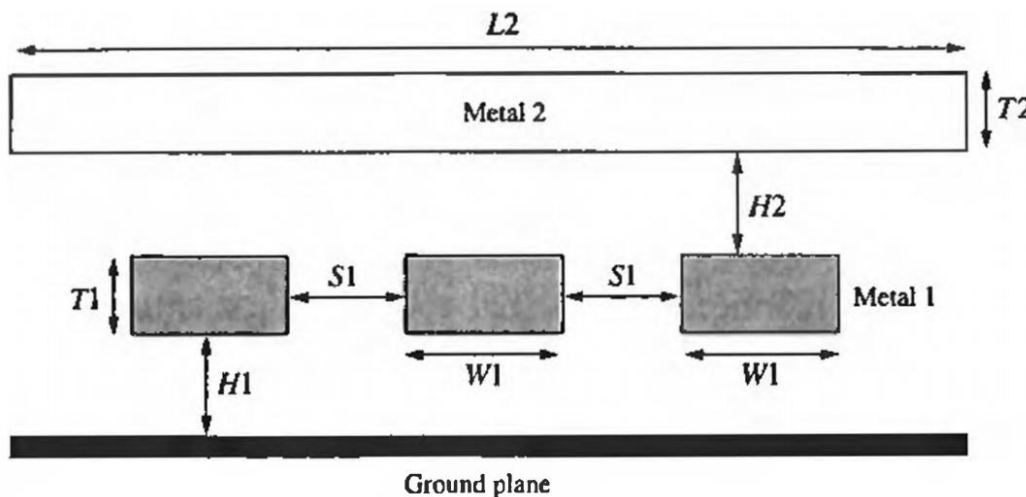
This distance between metals can be just 10's of nm!

B 

Capacitive Coupling

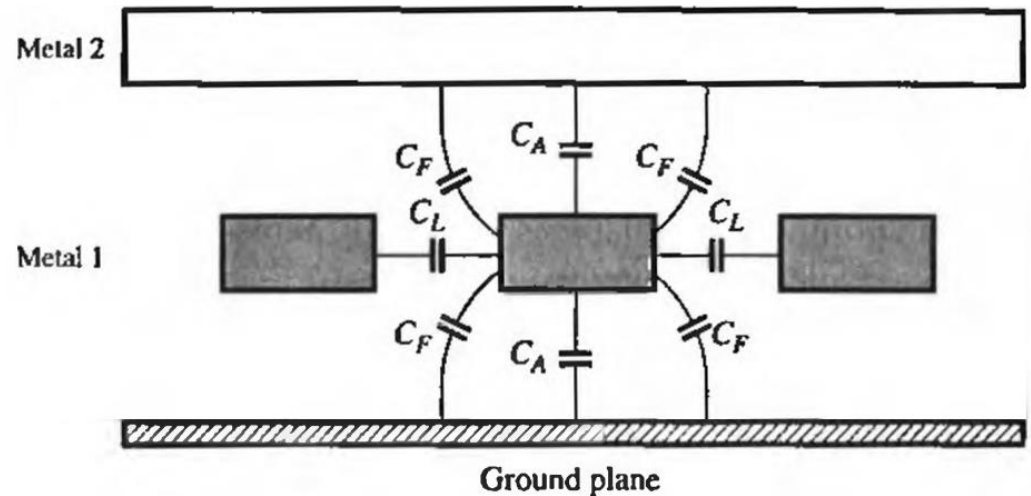
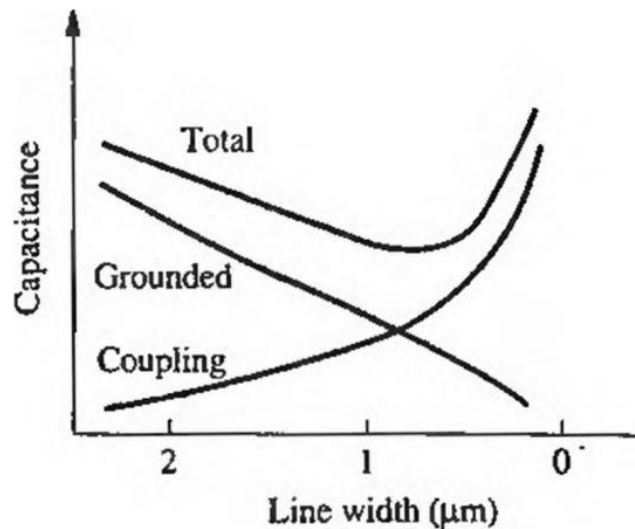
- Reduced spacing = higher coupling capacitance

$$C = \epsilon \frac{A}{d}$$



Capacitive Coupling

- Reduced line width (smaller process)
 - Reduces C_A
 - Increases C_L

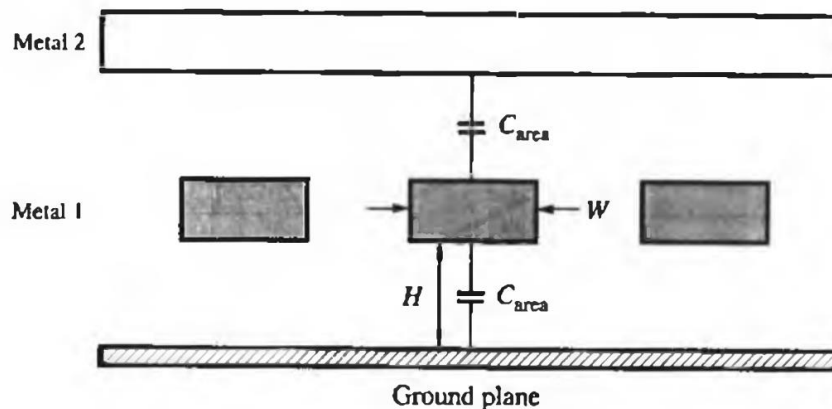


Area Capacitance

- Capacitance per unit length:
 - C_{area} often, but not always between signal and GND or V_{DD} .

$$C_{\text{area}} = \epsilon_{\text{ox}} \frac{W}{H}$$

$$C_{\text{area}} = \epsilon_{\text{ox}} \frac{W}{H} = 4\epsilon_0 \frac{W}{H} = 4(88.5 \times 10^{-4} \text{ fF}/\mu\text{m}) \frac{W}{H} = 0.035 \frac{W}{H} \text{ fF}/\mu\text{m}$$

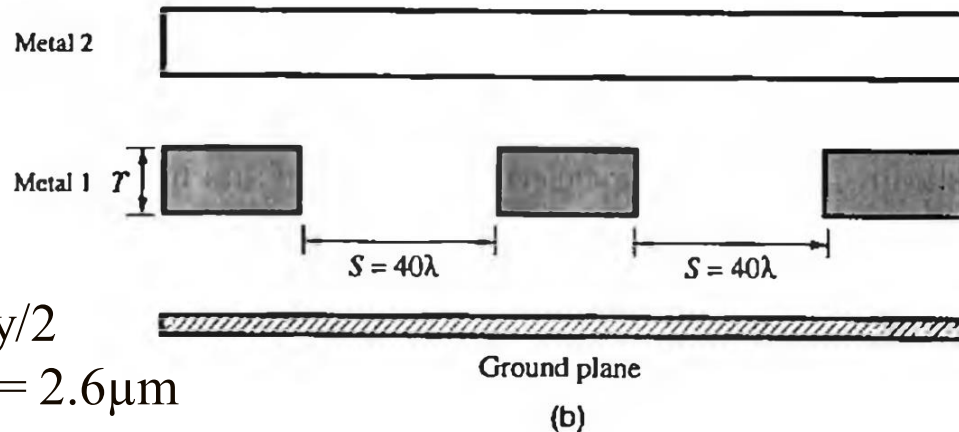


Lateral Capacitance

- Capacitance per unit length:
 - C_{lateral} usually between signals

$$C_{\text{lateral}} = \epsilon_{\text{ox}} \frac{T}{S}$$

$$C_{\text{lateral}} = \epsilon_{\text{ox}} \frac{T}{S} = 4\epsilon_0 \frac{T}{S} = 4(88.5 \times 10^{-4} \text{ fF}/\mu\text{m}) \frac{T}{S} = 0.035 \frac{T}{S} \text{ fF}/\mu\text{m}$$



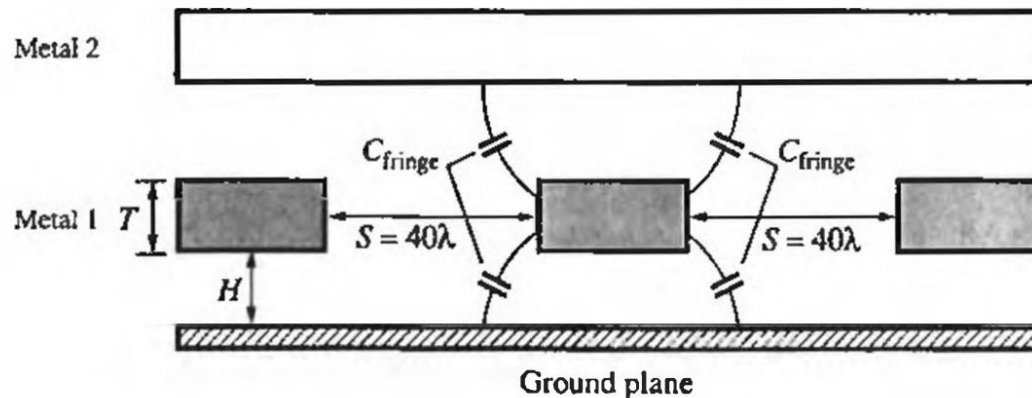
- $\lambda = \text{process technology}/2$
 - $40\lambda = 40 \cdot 130\text{nm}/2 = 2.6\mu\text{m}$

Fringe Capacitance

- Capacitance per unit length:
 - C_{fringe} usually between signal and GND or V_{DD} .

$$C_{\text{fringe}} = \epsilon_{\text{ox}} \ln\left(1 + \frac{T}{H}\right)$$

$$\begin{aligned} C_{\text{fringe}} &= \epsilon_{\text{ox}} \ln\left(1 + \frac{T}{H}\right) = 4(88.5 \times 10^{-4} \text{ fF}/\mu\text{m}) \ln\left(1 + \frac{0.5}{0.5}\right) \\ &= 0.025 \text{ fF}/\mu\text{m} \end{aligned}$$



Interconnect Capacitance

- Textbook ignores fringe capacitance
- You have been using the closely-spaced $C_{\text{int}} = 0.2 \text{ fF}/\mu\text{m}$

The total capacitance for the middle wire depends on whether the wires are closely spaced or widely spaced. The case for the closely spaced wires on the upper layers is given by

$$\begin{aligned}C_{\text{int}} &= 2C_{\text{area}} + 2C_{\text{lateral}} + 2C_{\text{fringe}} \\&\cong 2(0.03) + 2(0.070) + 2(0.0) \text{ fF}/\mu\text{m} \\&= 0.2 \text{ fF}/\mu\text{m}\end{aligned}$$

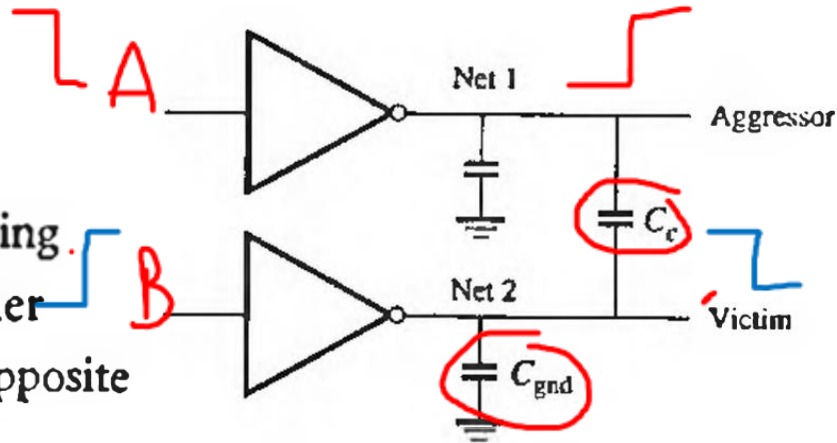
If the wires are widely spaced, we can use the following formulation:

$$\begin{aligned}C_{\text{int}} &= 2C_{\text{area}} + 2C_{\text{fringe}} + 2C_{\text{lateral}} \\&\cong 2(0.03) + 2(0.025) + 2(0.0) \text{ fF}/\mu\text{m} \\&\approx 0.1 \text{ fF}/\mu\text{m}\end{aligned}$$

Coupling Delay Effects

- C_L is input dependent

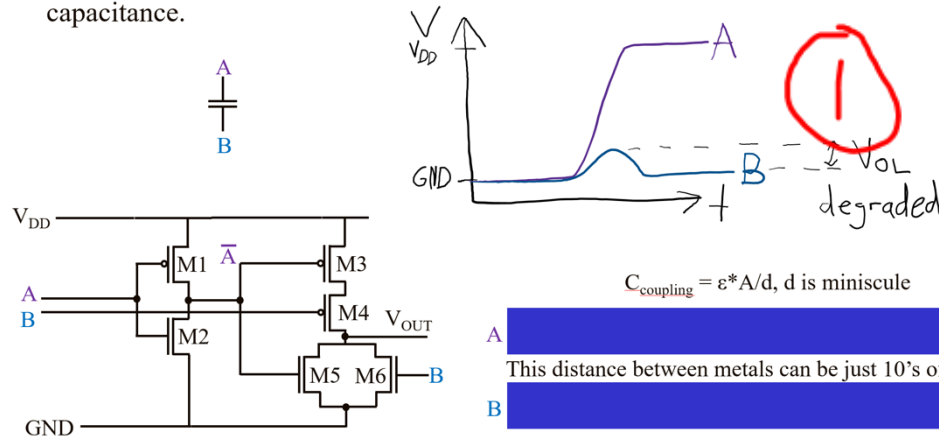
- ① $C_L = C_{gnd} + C_c$ victim not switching.
- $C_L = C_{gnd}$ both switching together
- ③ $C_L = C_{gnd} + 2C_c$ both switching opposite



$$\Delta V_{\text{victim}} = \frac{C_c \Delta V_{\text{aggressor}}}{C_{gnd} + C_c}$$

4 1

- If A and B run close together over a “long” distance, they will have a significant capacitance.



- B receives noise
 - ΔV_{victim} above
- A τ_p increased
 - C_L Above

Coupling Delay Effects

